Haniel Thomson

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Summary

Software Engineer building scalable and data-intensive AI systems. Built low latency backend infrastructure at Amazon Robotics supporting the Package Identification Technology Amazon unveiled at re:MARS 2022, the AI scanning systems that are now part of the largest warehouse robot fleet on earth. Built LLM agent pipelines automating manufacturing operations and RFQ processing for industrial customers. Current Ph.D. in AI student at Southwest Baptist University.

Experience

Software Engineer **Jan 2026 – Sep 2026**
Tarvo AI *Remote, USA*

- Increased sales by ∼180% across manufacturing companies by building an automated RFQ quoting platform (React, TypeScript, AWS, Postgres) processing 500+ requests per week with deterministic quote generation.
- Architected a fault-tolerant RFQ orchestration layer on Inngest and AWS Lambda with durable execution automatic retries, and concurrency limits, backed by a cron poller that claims unprocessed rows so no RFQ goes unprocessed.
- Built a multi-step LLM agent harness classifying inbound RFQs across CAD, PDF, DOC, spreadsheet, and image formats, extracting structured fields at 99.6% accuracy to automate quoting directly from customer email inboxes.
- Curated a 1000+ record golden dataset from manual review of RFQ mispredictions, filtering noisy examples and feeding few-shot examples back into the harness to improve extraction accuracy.
- Designed a deterministic graph based pricing engine letting customers define quoting logic from nodes (math formulas, data lookups, constants, variables) linked in a DAG, reducing customer onboarding from 3 hours to 10 minutes.
- Replaced manual AWS console provisioning with SST-based IaC across DynamoDB, S3, and Lambda, making environment setup reproducible and reducing setup time by ∼90%.

Software Engineer (Contract) **Dec 2024 – Aug 2025**
Amazon Robotics *Westborough, MA*

- Reduced smoke testing by ∼96.67% by building a concurrent low-latency microservice (Kotlin/Java, C++, Docker) to automate package identification testing with mock sensors and lock-free LMAX Disruptors, extended to 3 sister teams.
- Built a ETL release pipeline (AWS CDK, TypeScript) triggered by config or software changes, staging configs to S3 and software images to ECR, then deploying to edge devices via the fleet management system, saving 30 minutes per release.
- Optimized edge fleet provisioning by migrating Docker images to geo-proximate AWS regions (ECR) using AWS CDK, reducing provisioning time by 60%.
- Benchmarked system latency by designing bash-based workload simulations, collecting 160+ hours of performance data to identify and resolve critical bottlenecks and facilitate test automation.

Software Engineer Intern **Mar 2024 – Sep 2024**
Technical Consulting and Research, Inc. *Weston, CT*

- Owned the end-to-end backend architecture for a greenfield e-commerce platform (0→1), designing a scalable Django/Python schema to support multi-tenant inventory logic and 100+ active users.
- Designed and implemented a strict Role-Based Access Control (RBAC) system via Auth0, securing user data and ensuring compliance with enterprise security requirements.

Software Engineer Intern **Jun 2023 – Aug 2023**
Staples Inc. *Framingham, MA*

- Built a high-throughput reconciliation engine using Java and Spring Boot to audit cross-database consistency, improving data validation reliability by 70% during a critical legacy migration.
- Engineered concurrent batch processing pipelines (Spring Batch, Jenkins) to handle 500,000+ records with low latency, ensuring data integrity across distributed systems.

Skills

Languages: Python, TypeScript, JavaScript, Java, Kotlin, SQL (PostgreSQL, MySQL)
Backend & Frameworks: Spring Boot, Django, Node.js, Express, Next.js, RESTful APIs, Batch Processing
Infrastructure & AI: AWS (CDK, CloudFormation, S3, ECR), Docker, Git, CI/CD pipelines, LLM Agents, RAG

Education

Southwest Baptist University **Expected Sep 2030**
Ph.D. in Artificial Intelligence (Remote) *Bolivar, MO*

Boston University **Jan 2024**
Master of Science in Computer Science, Specialization in Data-Centric Computing *Boston, MA*

St. Joseph's College of Engineering **May 2022**
Bachelor of Engineering in Computer Science and Engineering *Tamil Nadu, India*